



Data Analyst Project Report

Bengaluru City · End-to-End Analytics

TOOLS USED

 Python	 MySQL	 Power BI
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KEY METRICS SNAPSHOT

Total Bookings	103,024
Successful Bookings	63,967 (62.09%)
Cancelled Bookings	84,590
Total Booking Value	₹35 Million
Date Range	01-07-2024 to 31-07-2024
City	Bengaluru, India

01 · PROJECT OVERVIEW

This end-to-end Data Analyst project simulates a real-world analytics workflow for OLA, India's leading ride-hailing platform. The dataset covers one full month of ride operations in Bengaluru with 1,00,000+ rows spanning all booking statuses, vehicle types, payment modes, cancellation reasons, and ratings.

Project Pipeline

STEP 1 Data Generation <i>ChatGPT prompt → 1L row CSV dataset</i>	STEP 2 Data Cleaning <i>Python (Pandas) in Jupyter Notebook</i>	STEP 3 EDA & SQL Analysis <i>MySQL — 10 key business queries</i>
STEP 4 Dashboard — Power BI <i>5 pages: Overall, Vehicle, Revenue, Cancellation, Ratings</i>	STEP 5 Insight Extraction <i>Business questions answered from visuals</i>	STEP 6 Documentation <i>This project report</i>

Dataset Specifications

Total Rows	~1,00,000 (1 Lakh)
Time Period	July 2024 (1 Month)
City	Bengaluru — 50 pickup/drop locations
Vehicle Types	7 (Prime Sedan, Prime SUV, Prime Plus, Mini, Auto, Bike, E-Bike)
Booking Status Split	Success 62% Driver Cancel 18% Customer Cancel 7%
Payment Methods	Cash, UPI, Credit Card, Debit Card
Incomplete Rides	< 6%

02 · DATA CLEANING (Python / Pandas)

The raw CSV dataset was cleaned and preprocessed using Python in a Jupyter Notebook. Below are the key data cleaning operations performed:

Cleaning Operations Performed

- **Removed duplicate Booking IDs**
- **Handled null values in Driver_Ratings and Customer_Rating (only for successful rides)**
- **Standardized date/time formats to YYYY-MM-DD and HH:MM:SS**
- **Validated Booking_Status values against expected categories**
- **Filled cancellation reason columns with 'N/A' for non-cancelled rides**
- **Ensured Incomplete_Rides_Reason is blank when Incomplete_Rides = 'No'**
- **Converted Booking_Value and Ride_Distance to numeric types**
- **Cross-validated cancellation rates against defined thresholds (customer < 7%, driver < 18%)**

Python Snippet — Data Quality Check

```
import pandas as pd

df = pd.read_csv('Data_SET_OLA.csv')

# Check nulls
print(df.isnull().sum())

# Validate booking status categories
print(df['Booking_Status'].value_counts())

# Cancellation rate check
cust_cancel = (df['Booking_Status']=='Cancelled by Customer').mean()
drv_cancel = (df['Booking_Status']=='Cancelled by Driver').mean()
print(f'Customer cancel rate: {cust_cancel:.2%}')
print(f'Driver cancel rate: {drv_cancel:.2%}')
```

03 - SQL EXPLORATORY DATA ANALYSIS

10 business questions were answered using MySQL. All queries were saved as views for efficient re-use in the Power BI dashboard.

Q1. Retrieve all successful bookings

```
SELECT * FROM bookings
WHERE Booking_Status = 'Success';
```

Q2. Find the average ride distance for each vehicle type

```
SELECT Vehicle_Type, AVG(Ride_Distance) AS avg_distance
FROM bookings
GROUP BY Vehicle_Type;
```

Q3. Get the total number of cancelled rides by customers

```
SELECT COUNT(*) AS total_customer_cancels
FROM bookings
WHERE Booking_Status = 'Cancelled by Customer';
```

Q4. Top 5 customers who booked the highest number of rides

```
SELECT Customer_ID, COUNT(Booking_ID) AS total_rides
FROM bookings
GROUP BY Customer_ID
ORDER BY total_rides DESC
LIMIT 5;
```

Q5. Rides cancelled by drivers due to personal & car-related issues

```
SELECT COUNT(*) AS personal_car_cancels
FROM bookings
WHERE Cancelled_Rides_by_Driver = 'Personal & Car related issue';
```

Q6. Max and min driver ratings for Prime Sedan

```
SELECT MAX(Driver_Ratings) AS max_rating,
       MIN(Driver_Ratings) AS min_rating
FROM bookings
WHERE Vehicle_Type = 'Prime Sedan';
```

Q7. All rides where payment was made via UPI

```
SELECT COUNT(*) AS upi_rides
FROM bookings
WHERE Payment_Method = 'UPI';
```

Q8. Average customer rating per vehicle type

```
SELECT Vehicle_Type, AVG(Customer_Rating) AS avg_customer_rating
```

```
FROM bookings  
GROUP BY Vehicle_Type;
```

Q9. Total booking value of successfully completed rides

```
SELECT SUM(Booking_Value) AS total_successful_value  
FROM bookings  
WHERE Booking_Status = 'Success';
```

Q10. List all incomplete rides with reason

```
SELECT Booking_ID, Incomplete_Rides_Reason  
FROM bookings  
WHERE Incomplete_Rides = 'Yes';
```

04 · POWER BI DASHBOARD

The Power BI report is structured into 5 dedicated pages, each targeting a specific analytical domain. Below are screenshots from each page with commentary.

Page 1 — Overall

High-level KPIs: Total Bookings (103,024), Total Booking Value (₹35M), Booking Status Breakdown pie chart, and Ride Volume Over Time trend line.

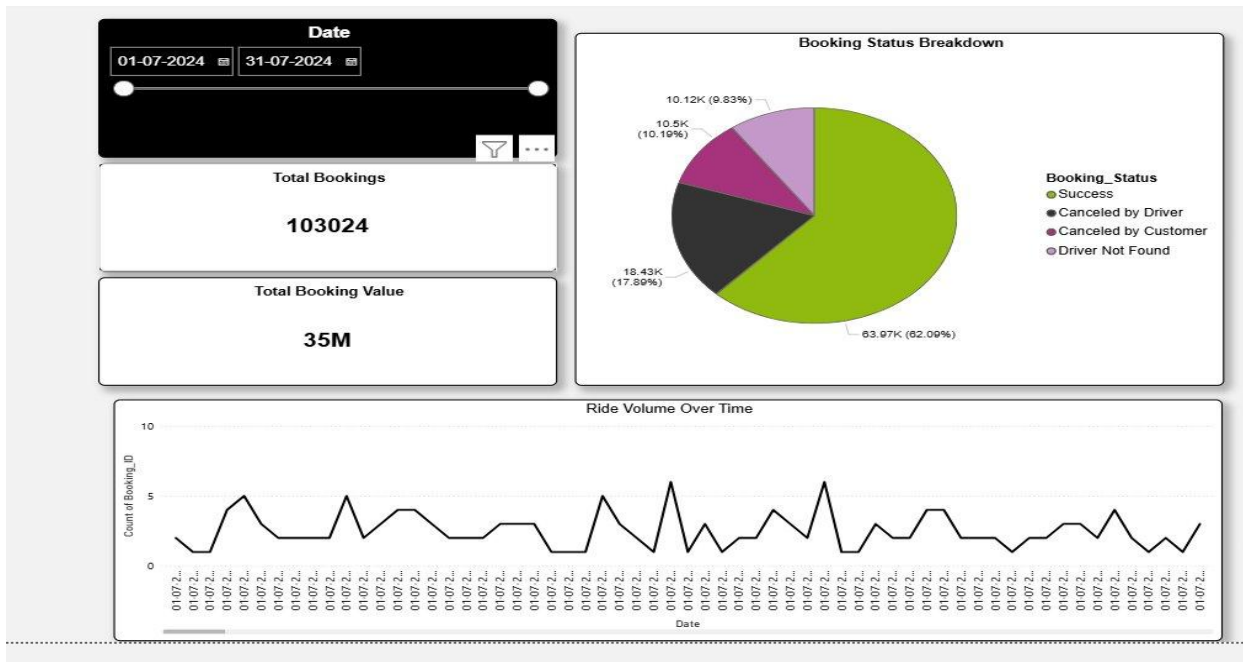


Figure 1: Overall Dashboard — Booking Status Breakdown & Ride Volume Over Time

Page 2 — Vehicle Type

Comparative performance of all 7 vehicle types across Total Booking Value, Successful Booking Value, Avg. Distance Travelled, and Total Distance Travelled.

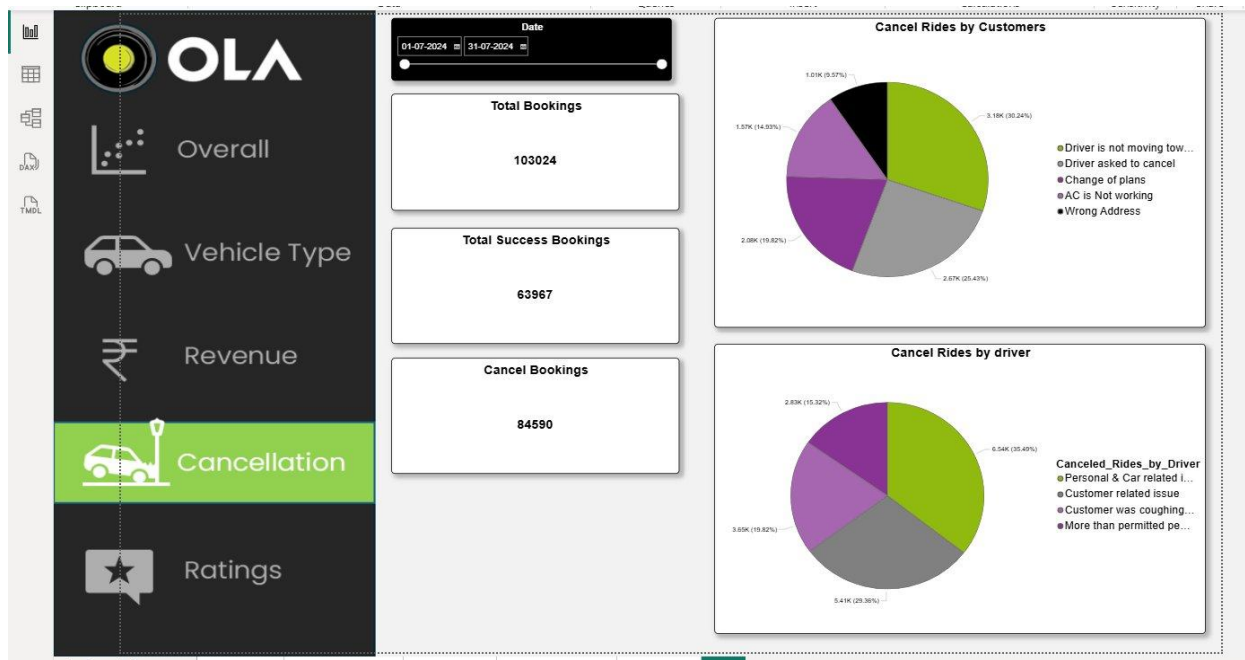


Figure 2: Vehicle Type Page — Performance comparison table

Page 3 — Revenue

Revenue by Payment Method (Cash dominates at ~₹20M, followed by UPI at ~₹14M), Top 5 Customers by Booking Value, and Ride Distance Distribution per day across July.

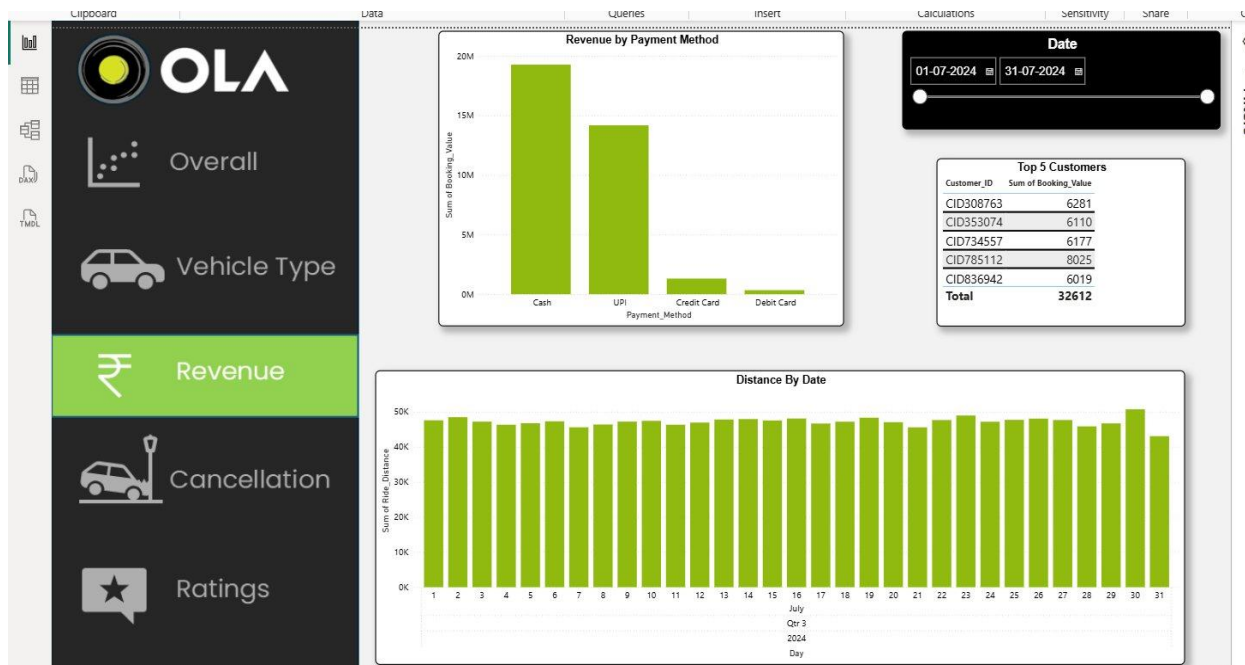


Figure 3: Revenue Page — Payment methods and distance trends

Page 4 — Cancellation

Cancel Bookings: 84,590 total. Two pie charts show reasons for customer cancellations (top: 'Driver not moving' 30.24%) and driver cancellations (top: 'Personal & Car related issue' 35.49%).

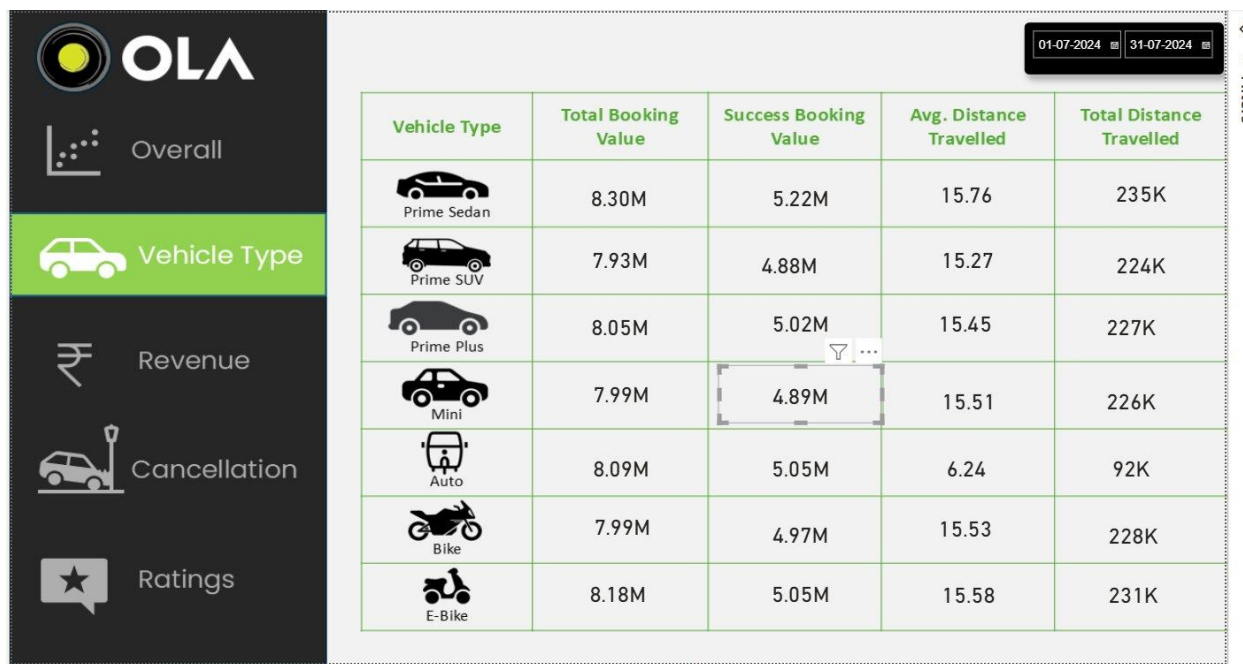


Figure 4: Cancellation Page — Reason breakdowns for customer and driver cancels

Page 5 — Ratings

Driver and Customer ratings shown per vehicle type. Ratings are consistently near 4.00 across all categories indicating a well-balanced service quality.

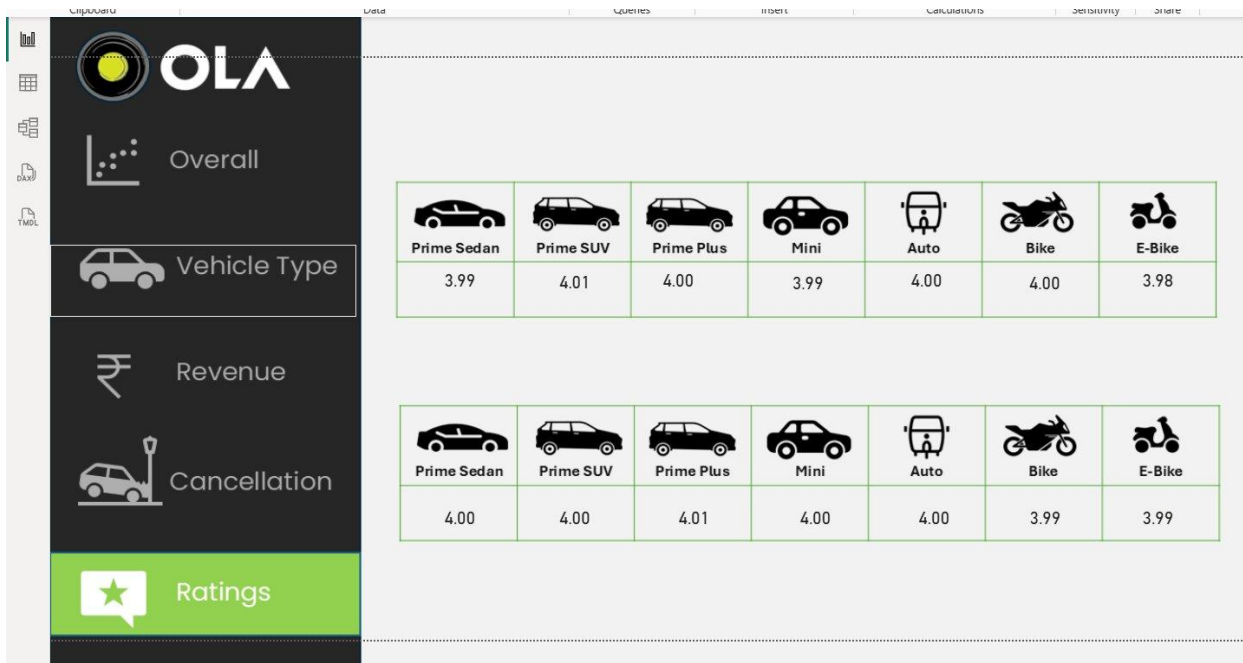


Figure 5: Ratings Page — Driver and Customer ratings by vehicle type

05 - VEHICLE TYPE ANALYSIS

Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance	Total Distance
Prime Sedan	₹8.30M	₹5.22M	15.76 km	235K km
Prime SUV	₹7.93M	₹4.88M	15.27 km	224K km
Prime Plus	₹8.05M	₹5.02M	15.45 km	227K km
Mini	₹7.99M	₹4.89M	15.51 km	226K km
Auto	₹8.09M	₹5.05M	6.24 km	92K km
Bike	₹7.99M	₹4.97M	15.53 km	228K km
E-Bike	₹8.18M	₹5.05M	15.58 km	231K km

Key Observations

- Prime Sedan leads in average ride distance at 15.76 km, indicating long-haul preference.
- Auto has the shortest average distance (6.24 km), typical for short intra-city trips.
- E-Bike has the highest Total Booking Value at ₹8.18M, showing strong eco-vehicle demand.
- All premium vehicles (Sedan, SUV, Plus) maintain consistent success booking values around ₹5M.

06 - RATINGS ANALYSIS

Vehicle Type	Driver Rating	Customer Rating
Prime Sedan	3.99	4.00
Prime SUV	4.01	4.00
Prime Plus	4.00	4.01
Mini	3.99	4.00
Auto	4.00	4.00
Bike	4.00	3.99
E-Bike	3.98	3.99

Key Observations

- • All vehicle types maintain driver and customer ratings tightly clustered between 3.98 – 4.01.
- • Prime Plus leads customer ratings at 4.01, reflecting superior ride comfort.
- • E-Bike and Bike show the lowest driver ratings (3.98–3.99), suggesting room for improvement.
- • High consistency in ratings signals a well-controlled service quality across OLA's fleet.

07 · CANCELLATION ANALYSIS

Total Cancellations	84,590
Cancelled by Customer	~10,500 (10.19%)
Cancelled by Driver	~18,430 (17.89%)
Driver Not Found	~10,120 (9.83%)

Top Customer Cancellation Reasons

Driver not moving towards pickup	30.24% — highest
Change of plans	25.43%
AC is not working	19.82%
Driver asked to cancel	14.93%
Wrong Address	9.57%

Top Driver Cancellation Reasons

Personal & Car related issues	35.49% — highest
Customer related issue	29.36%
Customer was coughing/sick	19.82%
More than permitted passengers	15.32%

08 · REVENUE ANALYSIS

Total Revenue (All Bookings)	₹35 Million
Revenue — Successful Rides	Highest contribution
Top Payment Method	Cash (~₹20M)
2nd Payment Method	UPI (~₹14M)
Top Customer (CID785112)	₹8,025

Top 5 Customers Combined	₹32,612
Consistent Daily Distance	~43K – 50K km/day

Revenue Insights

- Cash remains the dominant payment method, suggesting many riders prefer offline payments.
- UPI has strong traction at ~40% of cash volume — digital payment adoption is growing.
- Ride distances are highly consistent throughout July, indicating stable operational demand.
- The top 5 customers account for only ₹32K out of ₹35M — revenue is well-distributed.

09 - CONCLUSIONS & RECOMMENDATIONS

Business Conclusions

- 62% booking success rate is a solid operational baseline, but the 38% failure gap presents a major improvement opportunity.
- Driver cancellations (17.89%) are the single largest leakage point — reducing personal/car issues through vehicle maintenance programs could yield significant revenue gains.
- The 'driver not moving towards pickup' reason (30.24% of customer cancels) points to GPS or incentive alignment issues.
- Auto vehicles serve a distinct short-distance market (6.24 km avg.) — pricing and promotions should be calibrated differently from premium vehicles.
- Ratings near 4.0 across the board indicate consistent service quality, which is a strong trust signal.

Analyst Recommendations

- Implement driver location nudges and penalties for early-stage non-movement cancellations.
- Introduce cashless incentive programs to shift more payments to UPI/Cards, reducing cash handling risk.
- Use weekday/weekend demand data to introduce dynamic pricing models and driver surge bonuses.
- Create a dedicated Auto micro-segment analytics track given its distinct usage patterns.
- Set up real-time cancellation reason dashboards for operations teams to act within the same day.